**ReMis [A fun title could go here]**

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## Introduction

Structural equation models (SEMs) are widely used in applied research to test hypotheses about latent constructs and their relationships. In practice, replication is typically assessed by fitting the same model to a new dataset and evaluating whether global fit indices remain acceptable or whether selected parameters retain statistical significance. Empirically, these approaches often yield ambiguous or unstable outcomes. Models may exhibit acceptable global fit across datasets while differing substantially in parameter estimates, reflecting the non-uniqueness of the model-implied covariance structure. Conversely, parameters that were statistically significant in an original study may fail to reach significance in a replication due to differences in sample size, measurement quality, or sampling variability. In addition, estimation itself may be unstable. A model that converges in one dataset may fail to converge in another sample drawn from the same population, or even in resampled versions of the original data. Such behavior indicates sensitivity of the model to sampling fluctuations or weak identification, and implies that replication may fail at the level of model estimation rather than parameter comparison. These issues suggest that replication in SEM requires an explicit framework for distinguishing model fit, parameter stability, and theoretical stability. We argue that replication of SEMs can be assessed using the logic of invariance testing: measurement invariance establishes whether parameters can be meaningfully compared across datasets, parameter stability provides statistical evidence for replication, and theoretical stability determines whether the substantive claim encoded by the model is preserved.

## Replication

Replication is the ability to obtain consistent findings when a study is repeated with new independent data being collected under the same or sufficiently similar methodological and theoretical conditions as the original study. In the context of empirical research, a finding can be understood at different analytical levels: as an empirical regularity, as an estimand-based claim, or as a theoretical claim about an underlying mechanism that generates observed patterns. The type or extent of replication relevant to a finding depends on which of these levels the finding is defined at.

Radder (1992) characterizes replication as comprising three distinct forms: (a) material replication; (b) reproducibility of the theoretical interpretation; and (c) replication of the experimental result through an alternative material procedure. Schmidt (2009) builds on this definition and establishes the concepts of direct replication as a restrictive and narrow form of replication and conceptual replication as a wider notion of replication.

Direct replication refers to a repetition of the original experimental or observational procedure aimed at re-estimating the same estimand under equivalent conditions. Formally, direct replication concerns approximate agreement in the estimation of the same substantive quantity:

$$\theta^{*(1)} \approx \theta^{*(2)}.$$

Under finite sampling, exact numerical equality of parameter estimates is generally implausible. Thus, direct replication refers to a repetition of the original experimental or observational procedure aimed at re-estimating the same estimand under equivalent conditions. This in turn relates to non-experimental methods as when a new study estimates the same estimand under conditions that are equal to the original, operationalizing the variables in the same way, with differences limited to incidental sources of variation such as sample, time, or setting. Under this definition, deviations from the original estimate should be small, but exact numerical equality is not required.

By contrast, conceptual replication concerns the stability of a theoretical claim under changes in operationalization. This relates to Radder (1992), his b and c compartments of a replication, so that the same theoretical claim or construct is supported using a different estimand that is theoretically analogous to the original. This involves changes in operationalization, such as different independent or dependent measures, while retaining the underlying conceptual target. Conceptual replication can then be expressed through a mapping $\phi$ such that

$$C(\theta^{(1)}) = C(\phi(\theta^{(2)})),$$

where $C(\cdot)$ denotes the theoretical claim supported by the estimand. In this case, replication is evaluated at the level of theoretical equivalence rather than parameter identity.

## Replicating Structural Equation Models

According to Kaplan (2009), SEM comprises a class of methodologies designed to express a hypothesized model through summary statistics derived from empirical observations, parameterized by a reduced set of structural parameters. As SEMs are also considered to be a way to formulate social science theories(De Leeuw and Berk 2009), for a SEM to be deemed replicated, the encompassed theory needs to hold in the data, either globally or regarding certain parameters.

For models involving continuous latent variables, the associated hypotheses typically pertain to the means, variances, and covariances of the observed variables. Hypotheses may also concern effect sizes, ordering of parameter magnitudes, or specific parameter values. Because SEMs often contain many estimated parameters, replication should generally focus on those outcomes that are substantively relevant to the hypothesis under study.

### ***Global model fit versus parameter-level replication***

Global model fit concerns the adequacy of the model-implied covariance structure. Let $\Sigma(\theta)$ denote the covariance matrix implied by the model parameters $\theta$, and let $\mathbf{S}^{(d)}$ denote the observed covariance matrix in dataset $d$. Global fit evaluates whether

$$\Sigma(\theta) \approx \mathbf{S}.$$

However, good global fit in both datasets does not imply replication at the level of parameters. Multiple parameter configurations may produce similar model-implied covariance structures. Global fit therefore concerns the adequacy of the model-implied distribution, whereas replication at the level of parameters concerns the stability of theoretically relevant estimands.

Accordingly, replication in SEM should be evaluated with respect to those parameters that directly encode the substantive claim under investigation. Let $\theta^*$ denote the subset of theory-relevant model parameters. Figure X illustrates this idea. The highlighted paths represent theory-relevant relations whose stability is relevant for replication assessment, whereas the remaining model parameters are not necessarily central to the substantive hypothesis under investigation.

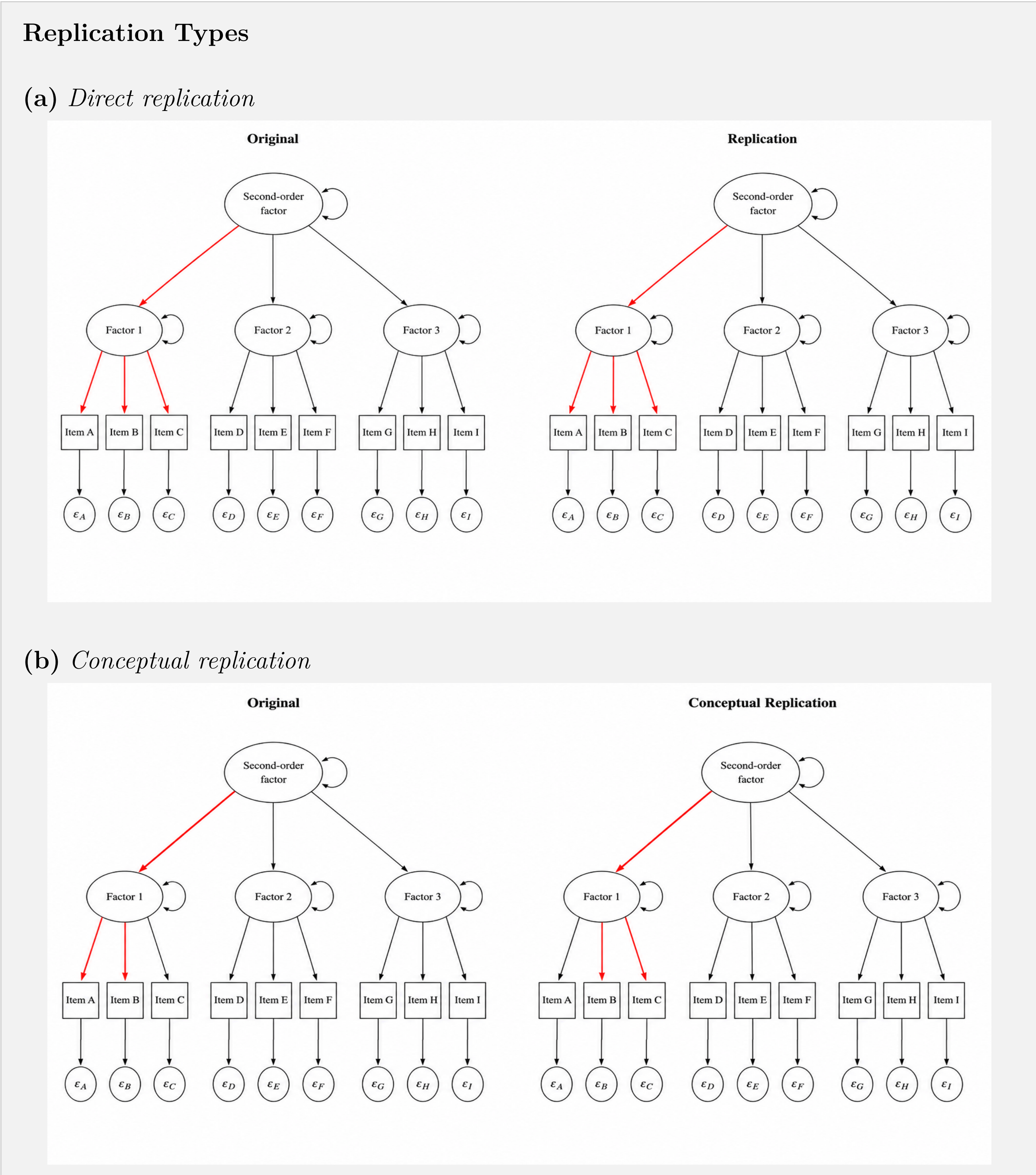


**Figure 1**

*Note. Structural equation model representations of direct and conceptual replication. Highlighted paths indicate theory-relevant parameters ($\theta^*$) whose stability is relevant for replication assessment. In direct replication (a), the same theory-relevant relations are evaluated across studies. In conceptual replication (b), different parameter configurations may support the same higher-level theoretical claim.*

Replication at the parameter level may then be expressed as

$$\theta^{*(1)} \approx \theta^{*(2)}.$$

This formulation captures stability of theory-relevant estimands rather than merely adequacy of the model-implied covariance structure. At the same time, parameter stability should not be equated with replication success itself. Replication ultimately concerns the stability of the substantive conclusions supported by these parameters.

If replication is viewed as a comparison of theory-relevant parameters across independent datasets, both datasets may be treated as realizations of the same underlying population model. The resulting problem is therefore one of systematic parameter comparison across datasets. In the following sections, we argue that the logic of measurement and structural invariance provides a natural methodological framework for addressing this problem.

## Invariance Testing

Measurement invariance (MI) refers to the statistical property that a measurement instrument assesses the same latent construct equivalently across different groups. In other words, the measurement model operates in the same manner across groups. When MI is established, meaningful comparisons of the underlying construct between groups are justified.

From a statistical modelling perspective, MI refers to invariance of the measurement parameters across groups.

$$\theta_M^{(g)} = \theta_M \quad \forall g,$$

where $\theta_M$ denotes the set of measurement parameters, including factor loadings, intercepts, and residual variances.

Structural invariance (SI), by contrast, refers to equality of structural coefficients across groups, that is, to the requirement that the same association between latent factors exists across groups.

$$\theta_S^{(g)} = \theta_S \quad \forall g,$$

where $\theta_S$ denotes the set of structural parameters governing the relationships among latent variables.

Traditionally, MI and SI are characterized via increasingly strict levels of invariance that each add further restrictions on statistical parameters across groups. In the classical view, MI is attained only when configural, metric, scalar, and residual invariance hold. These correspond to successive equality restrictions on factor loadings, intercepts, and residual variances across groups. More recent perspectives conceptualize invariance more generally as the absence of systematic moderation of model parameters by variables external to the model. Formally, if model parameters depend on group membership through

$$\theta^{(g)} = \theta + \Delta^{(g)},$$

then invariance corresponds to the condition

$$\Delta^{(g)} = 0 \quad \forall g.$$

Covariance invariance analyses evaluate whether covariances among latent factors are equivalent across groups, whereas SI analyses examine whether the structural relationships among latent factors are equal across groups. Together, these analyses assess whether relationships among latent variables remain stable across groups or vary systematically as a function of group membership.

**Using Invariance Testing to replicate SEMs**

The central argument of this paper is that the logic of invariance testing can be repurposed as a methodological framework for assessing replication in SEM. Measurement invariance is not itself a theory of replication. Rather, it provides a statistical procedure for evaluating whether theory-relevant parameters remain stable under a specified source of variation. In traditional applications, this variation is induced by group membership. In replication research, the relevant source of variation is independent data collection. The methodological role of invariance testing therefore remains unchanged, while the substantive interpretation shifts from group comparison to replication assessment.

Whereas measurement invariance is conditional on group definitions, replication is conditional on independent datasets. Nevertheless, both involve the assessment of parameter stability under variation. This makes the relation between replication and invariance testing explicit: both evaluate whether theory-relevant parameters remain stable when conditions change, differing primarily in the source of that variation.

Replication of SEMs across an original and a replication sample should therefore proceed within a sequential multiple-group manner.

Replication, as defined above, refers to the stability of theory-relevant parameters across independent datasets and is not conditional on measurement invariance. However, in the context of SEM, the empirical assessment of structural parameter stability requires that the underlying measurement model operates equivalently across datasets. Without such equivalence, observed differences in structural parameters may reflect differences in measurement rather than differences in the underlying relations of interest. Measurement invariance is therefore a necessary condition for the valid interpretation of structural replication.

The analogy to invariance testing applies primarily to the statistical assessment of parameter stability. As in measurement invariance research, evidence against exact equality does not necessarily imply substantively meaningful non-equivalence. Accordingly, the outcome of invariance tests should be interpreted with respect to the theoretical claims represented by the constrained parameters. This suggests a distinction between the statistical assessment of replication and its substantive interpretation. Measurement invariance provides the methodological framework, parameter stability constitutes the statistical target, and theoretical stability serves as the criterion for replication. The former concerns the stability of theory-relevant parameters across datasets, whereas the latter concerns whether any observed differences alter the theoretical conclusions supported by the model.

First, measurement invariance between the two samples has to be established:

$$\theta_M^{(1)} = \theta_M^{(2)}.$$

At a minimum, sufficient measurement invariance must be established for the structural parameters of interest to be meaningfully compared across datasets. Conditional on sufficient measurement invariance, structural replication is assessed by comparing an unconstrained multiple-group model, in which the structural parameters of interest are freely estimated across datasets, with a constrained model in which they are set to be equal:

$$H_0 : \theta_S^{*(1)} = \theta_S^{*(2)}.$$

This comparison could be evaluated with a likelihood-ratio statistic. If a model in which the parameters are constrained to be equal across groups fits the data no worse than a model in which they are freely estimated, this is consistent with structural stability of those parameters. Conversely, if the constrained model fits the data worse, this indicates that the parameters differ across groups. At the same time, such tests should be interpreted cautiously because they are sensitive to sample size and do not by themselves capture uncertainty in a substantively calibrated way.

**Ambiguous Replication**

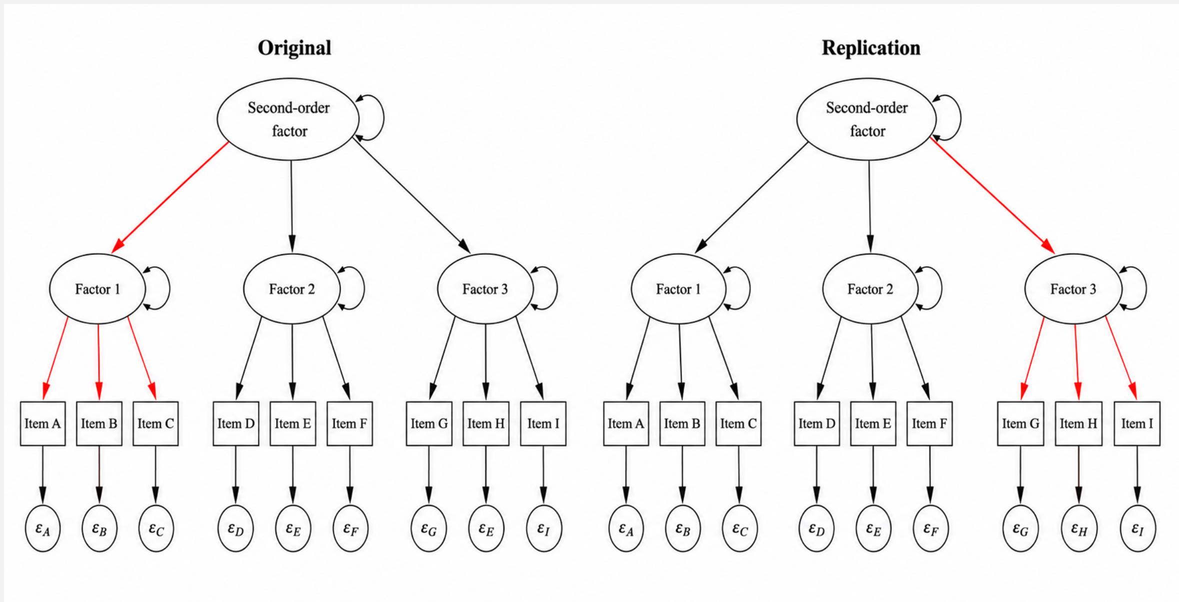


**Figure 2**

*Note. Ambiguous replication outcome in SEM. Differences are observed in theory-relevant parameters across studies, but their implications for replication remain unclear. Evaluation requires consideration of measurement equivalence and the extent to which the observed discrepancies alter the underlying theoretical claim.*

## Quantifying and Evaluating Discrepancies

Evidence against exact equality does not necessarily imply replication failure. As in the measurement invariance literature, statistically detectable differences may have negligible substantive consequences [NYE Reference]. The question therefore shifts from whether parameters differ to whether the observed differences alter the theoretical claim supported by the model.

This perspective is consistent with the earlier formulation of conceptual replication, where replication was evaluated through preservation of a theoretical claim rather than equality of estimands. The same logic can be extended to direct replication. Let

$$C(\theta)$$

denote the theoretical claim implied by a parameter configuration $\theta$. Direct replication may then be interpreted as preservation of the same theoretical claim across independent datasets:

$$C\left(\theta^{*(1)}\right) = C\left(\theta^{*(2)}\right).$$

Under this formulation, parameter stability constitutes evidence for replication, but replication itself is evaluated at the level of theoretical conclusions. Consequently, statistically detectable parameter differences need not imply replication failure if the substantive claim represented by those parameters remains unchanged.

This formulation unifies direct and conceptual replication within a common framework. For conceptual replication, preservation of a theoretical claim is evaluated across theoretically related estimands:

$$C\left(\theta^{(1)}\right) = C\left(\phi\left(\theta^{(2)}\right)\right),$$

where $\phi$ denotes a mapping between different operationalizations. By contrast, direct replication concerns preservation of the same theoretical claim for the same estimand across independent datasets:

$$C\left(\theta^{*(1)}\right) = C\left(\theta^{*(2)}\right).$$

This implies that replication of SEMs is best understood as a graded, parameter- and claim-relative process.

Because theoretical claims are evaluated through model parameters, assessment of theoretical stability requires a corresponding assessment of parameter stability. Let

$$\theta^* = \{\theta_1^*, \dots, \theta_r^*\}$$

denote the set of parameters relevant to the theoretical claim of interest. Parameter stability can then be represented as the vector

$$\hat{S} = \left(\mathbb{I}\left(\left|\hat{\theta}_j^{*(1)} - \hat{\theta}_j^{*(2)}\right| \leq \epsilon_j\right)\right)_{j=1}^{r},$$

where $\mathbb{I}(\cdot)$ is an indicator function and $\epsilon_j$ is a substantively justified tolerance for parameter $j$. A parameter is said to be empirically *stable* if

$$\left|\hat{\theta}_j^{*(1)} - \hat{\theta}_j^{*(2)}\right| \leq \epsilon_j.$$

Direct and conceptual replication therefore differ in the degree of operational similarity between studies, but both may be understood as instances of theoretical stability.

In direct replication, stability is assessed for the same estimand across datasets, whereas in conceptual replication stability is assessed across theoretically related estimands linked through $\phi$.

Full replication corresponds to preservation of all theory-relevant claims across datasets. Partial replication corresponds to preservation of only a subset of those claims. Parameter stability, as represented by the criterion above, provides empirical evidence regarding replication but should not be interpreted as identical to replication itself. This perspective renders approximate and partial replication theoretically grounded rather than ad hoc, because limited deviations in parameter estimates can be interpreted as meaningful quantitative differences rather than categorical failures of replication, paralleling contemporary approaches to approximate invariance.

## Practical Evaluation of Replication Outcomes

The preceding sections established a distinction between parameter stability and theoretical stability. Measurement invariance provides the methodological framework for comparing parameters across datasets, whereas replication is ultimately evaluated with respect to the theoretical claims supported by those parameters. The remaining question is therefore how observed parameter discrepancies should be quantified and interpreted in practice.

Consistent with Nye et al.'s consequence-based perspective on measurement nonequivalence, replication assessment may focus on the substantive consequences of parameter discrepancies rather than their mere statistical presence. In practical applications, replication assessment therefore proceeds in three stages: (a) establish measurement comparability, (b) assess parameter stability, and (c) evaluate whether any remaining discrepancies alter the theoretical claim of interest. The framework proposed here does not prescribe a unique discrepancy metric. Rather, the choice of metric should reflect the substantive consequences of the observed parameter differences. In this sense, replication assessment may be viewed as consequence-based: the relevant question is not merely how much parameter estimates differ, but how much the theoretical implications of the model change as a result.

Several operationalizations are possible. Discrepancies may be expressed as standardized parameter differences, relative deviations from the original estimate, changes in latent covariance structures, or changes in the model-implied covariance matrix. The common principle is that discrepancy measures should be interpreted with respect to their consequences for the theoretical claim under investigation.

### *Example 1: Full Replication Despite Parameter Differences*

Consider a structural path relating a latent predictor to a latent outcome. In the original study, the standardized path coefficient is

$$\widehat{\beta}^{(1)} = .50,$$

whereas the replication study yields

$$\widehat{\beta}^{(2)} = .45.$$

The absolute discrepancy is

$$\left|\widehat{\beta}^{(1)} - \widehat{\beta}^{(2)}\right| = .05.$$

A relative discrepancy measure may be defined as

$$W = \frac{\left|\widehat{\beta}^{(1)} - \widehat{\beta}^{(2)}\right|}{\left|\widehat{\beta}^{(1)}\right|}.$$

Substituting the observed values gives

$$W = \frac{.05}{.50} = .10.$$

Thus, the replication estimate differs by approximately 10% from the original estimate. The discrepancy is therefore small relative to the magnitude of the original effect. Despite this discrepancy, both studies support the same substantive conclusion: the predictor exhibits a positive and moderately sized effect on the outcome. Under the framework proposed here, replication would therefore be considered successful because the theoretical claim is preserved:

$$C\left(\theta^{*(1)}\right) = C\left(\theta^{*(2)}\right).$$

***Example 2: Approximate Replication***

Now consider a smaller original effect

$$\widehat{\beta}^{(1)} = .10,$$

with the replication estimate

$$\hat{\beta}^{(2)} = .05.$$

The absolute discrepancy is again .05, but the relative discrepancy becomes

$$W = \frac{.05}{.10} = .50.$$

Thus, approximately half of the original effect has disappeared. The absolute discrepancy is identical to that in Example 1, yet its substantive consequences are considerably larger. This illustrates why absolute parameter differences alone may be misleading as indicators of replication success. Depending on the theoretical context, the underlying claim may still be regarded as preserved, but with weaker empirical support. This situation illustrates approximate replication: parameter differences exist, yet the theoretical implication remains largely intact.

***Example 3: Partial Replication***

Suppose a theory predicts three relations among latent variables.

| Relation | Study 1 | Study 2 |
|---|---|---|
| A $\rightarrow$ B | .70 | .68 |
| A $\rightarrow$ C | .40 | .41 |
| B $\rightarrow$ C | .50 | .20 |

A discrepancy measure based on the latent relation matrix may be defined as

$$W = \max \left| R^{(1)} - R^{(2)} \right|.$$

For the example above $W = .30.$, indicating that at least one theoretically relevant relation differs substantially across studies. Consequently, some theoretical claims are preserved while others are not. Under the proposed framework, this outcome would be interpreted as partial replication rather than complete success or complete failure.

### *Example 4: Replication Failure*

Finally, Consider an original study yielding $\hat{\beta}^{(1)} = .45$, whereas the replication study yields $\hat{\beta}^{(2)} = -.05$. Although both estimates may be statistically distinguishable from zero, the substantive interpretation changes fundamentally. The original study supports a positive relation between the constructs, whereas the replication study suggests no meaningful relation or even a weak negative association.

In this case, $C\left(\theta^{*(1)}\right) \neq C\left(\theta^{*(2)}\right)$, because the theoretical conclusion implied by the model is no longer preserved. Replication would therefore be considered unsuccessful.

These examples illustrate that replication assessment depends not only on the magnitude of parameter differences but also on their consequences for the theoretical claims represented by the model. Consequently, discrepancy measures should be interpreted with respect to the substantive conclusions they support rather than in isolation.

## Discussion

It follows that models in their entirety should not be treated as the primary objects of replication. Rather, models serve as entry points for evaluating the stability of the theoretical commitments they encode. This shifts attention away from global model fit toward the robustness of substantively meaningful parameters. Because parameter stability under model misspecification may reflect shared misspecification rather than stability of the underlying relations, global fit is a necessary but not sufficient condition for meaningful interpretation of replication. At the same time, imperfect global fit does not necessarily invalidate replication at the level of specific theory-relevant parameters. Measurement quality also becomes central to replication validity, because instability across datasets may reflect violations of measurement equivalence rather than instability of the structural relations themselves. Replication claims that do not explicitly address measurement equivalence therefore remain ambiguous.

Conceptual replication has a formal structure under this perspective as it concerns stability of higher-level theoretical claims under deliberate changes in operationalization. This can be understood as stability of $C(\cdot)$ under transformations of the estimand space, rather than equality of the same parameter across datasets. Replication success therefore becomes explicitly conditional on the estimand, the theoretical scope, and the operational context, requiring precise statements about what exactly has or has not replicated. The distinction between invariance testing and replication may therefore be understood as one of scope rather than kind. Invariance evaluates stability across groups within a dataset, whereas replication evaluates stability across independent datasets or study instantiations, but both address the same underlying question of robustness under variation.

## References


De Leeuw, J., and R. Berk. 2009. "Preface." In *Structural Equation Modeling: Foundations and Extensions*, 2nd ed., edited by J. De Leeuw and R. Berk. SAGE Publications, Inc. https://doi.org/10.4135/9781452226576.

Kaplan, David. 2009. *Structural Equation Modeling (2nd Ed.): Foundations and Extensions*. 2nd ed. Edited by J. De Leeuw and R. Berk. SAGE Publications, Inc. https://doi.org/10.4135/9781452226576.

Radder, Hans. 1992. "Experimental Reproducibility and the Experimenters' Regress." *PSA: Proceedings of the Biennial Meeting of the Philosophy of Science Association* 1992: 63–73. http://www.jstor.org/stable/192744.

Schmidt, Stefan. 2009. "Shall We Really Do It Again? The Powerful Concept of Replication Is Neglected in the Social Sciences." *Review of General Psychology* 13 (2): 90–100. https://doi.org/10.1037/a0015108.